torch.corrcoef#

https://pytorch.org/docs/stable/generated/torch.corrcoef.html#torch.corrcoef

Description:

Estimates the Pearson product-moment correlation coefficient matrix of the variables given by the input matrix, where rows are the variables and columns are the observations. The correlation coefficient matrix R is computed using the covariance matrix C as given by Rij=CijCii∗CjjR_{ij} = \frac{ C_{ij} } { \sqrt{ C_{ii} * C_{jj} } }Rij=Cii∗CjjCij Due to floating point rounding, the resulting array may not be Hermitian and its diagonal elements may not be 1. The real and imaginary values are clipped to the interval [-1, 1] in an attempt to improve this situation.

Function Signature

```
torch.corrcoef(input) → Tensor#
Parameters
Returns
```

Returns:

(Tensor) The correlation coefficient matrix of the variables.

Code Examples

```
>>> x = torch.tensor([[0, 1, 2], [2, 1, 0]])
>>> torch.corrcoef(x)
tensor([[ 1., -1.],
        [-1.,  1.]])
>>> x = torch.randn(2, 4)
>>> x
tensor([[-0.2678, -0.0908, -0.3766,  0.2780],
        [-0.5812,  0.1535,  0.2387,  0.2350]])
>>> torch.corrcoef(x)
tensor([[1.0000, 0.3582],
        [0.3582, 1.0000]])
>>> torch.corrcoef(x[0])
tensor(1.)
```

Notes & Warnings

NOTE

Note The correlation coefficient matrix R is computed using the covariance matrix C as given by Rij=CijCii∗CjjR_{ij} = \frac{ C_{ij} } { \sqrt{ C_{ii} * C_{jj} } }Rij=Cii∗CjjCij

NOTE

Note Due to floating point rounding, the resulting array may not be Hermitian and its diagonal elements may not be 1. The real and imaginary values are clipped to the interval [-1, 1] in an attempt to improve this situation.

NOTE

See also torch.cov() covariance matrix.

Detailed Documentation

Documentation

Estimates the Pearson product-moment correlation coefficient matrix of the variables given by the input matrix, where rows are the variables and columns are the observations.

The correlation coefficient matrix R is computed using the covariance matrix C as given by Rij=CijCii∗CjjR_{ij} = \frac{ C_{ij} } { \sqrt{ C_{ii} * C_{jj} } }Rij=Cii∗CjjCij

Due to floating point rounding, the resulting array may not be Hermitian and its diagonal elements may not be 1. The real and imaginary values are clipped to the interval [-1, 1] in an attempt to improve this situation.

input (Tensor) – A 2D matrix containing multiple variables and observations, or a Scalar or 1D vector representing a single variable.

(Tensor) The correlation coefficient matrix of the variables.

torch.cov() covariance matrix.